

Name of the Programme:	BCA (H)	
Title of the Course:	Set Theory	
Course Code	CS1885	
Semester:	II	
Name(s) of the Course Instructor(s)	Prof. Subhadip Ghosh	
Course credit	No. of hours per week (L + T + P)	Total no. of Teaching hours
2	1 + 1 + 0	30

Session	Module No.	Topic	Pedagogy /activities	Pre-reading/ Reference
1	1	Introduction to sets	Lecture + PPT/Activity	GS Rao, Ch. 1, pp. 1–10
2	1	Set-builder notation	Lecture + chalk and Talk/ PPT/Activity	GS Rao, Ch. 1, pp. 1–10
3	1	Types of sets (finite, infinite, singleton, null, equal, subset, universal)	Discussion+ Example	GS Rao, Ch. 2, pp. 31–45
4	1	Types of sets (finite, infinite, singleton, null, equal, subset, universal)	Discussion+ Example	GS Rao, Ch. 2, pp. 31–45
5	1	Cardinality	Lecture + chalk and Talk/ PPT/Activity	GS Rao, Ch. 3, pp. 83–96
6	1	Cardinality	Discussion+ Example	GS Rao Ch. 3, pp. 83–96
7	2	Operations on sets: Union,	Lecture + chalk and Talk/	GS Rao, Ch. 4, pp. 101–112

			PPT/Activity	
8	2	Intersection, Difference	Discussion+ Example	GS Rao, Ch. 4, pp. 113–125
9	2	Complement, Symmetric Difference	Lecture + chalk and Talk/ PPT/Activity	GS Rao, Ch. 4, pp. 126–145
10	2	Properties and laws of set operations	Discussion+ Example	GS Rao Ch. 4, pp. 126–145
11	2	Venn diagrams	Lecture + chalk and Talk/ PPT/Activity	GS Rao, Ch. 4, pp. 146–158
12	2	Cartesian product of sets	Discussion+ Example	GS Rao, Ch. 4, pp. 146–158
13	3	Definition of relation	Lecture + chalk and Talk/ PPT/Activity	GS Rao Ch. 5, pp. 165–178
14	3	Types of relations: Reflexive, Symmetric	Discussion+ Example	GS Rao, Ch. 5, pp. 165–178
15	3	Transitive, Equivalence	Discussion + Examples/ Activity	GS Rao, Ch. 5, pp. 179–198
16	3	Representation of relations	Discussion+ Example	GS Rao, Ch. 5, pp. 179–198
17	3	Matrix and digraph representation.	Lecture + chalk and Talk/ PPT/Activity	GS Rao, Ch. 5, pp. 199–210

18	3	Matrix and digraph representation	Discussion+ Example	GS Rao, Ch. 5, pp. 199–210
19	4	Definition of functions	Lecture + chalk and Talk/ PPT/Activity	GS Rao, Ch. 6, pp. 217–228
20	4	One-to-one and onto functions	Discussion+ Example	GS Rao, Ch. 6, pp. 217–228
21	4	Function composition	Lecture + chalk and Talk/ PPT/Activity	GS Rao, Ch. 6, pp. 229–242
22	4	Invertibility and inverse of functions	Discussion+ Example	GS Rao, Ch. 6, pp. 229–242
23	4	Pigeonhole Principle	Discussion+ Example	GS Rao, Ch. 6, pp. 243–275
24	4	Pigeonhole Principle	Discussion+ Example	GS Rao, Ch. 6, pp. 243–275
25	5	Rule of Sum and Product	Discussion+ Example	GS Rao, Ch. 7, pp. 281–292
26	5	Permutations and Combinations	Discussion+ Example	GS Rao, Ch. 7, pp. 281–292
27	5	Binomial theorem	Discussion+ Example	GS Rao, Ch. 7, pp. 293–305
28	5	Combination with repetition	Discussion+ Example	GS Rao, Ch. 7, pp. 293–305

29	5	Combination with repetition.	Discussion+ Example	GS Rao, Ch. 7, pp. 306–318
30	5	Combination with repetition.	Discussion+ Example	GS Rao, Ch. 7, pp. 306–318

Text Books	
1	G. Shanker Rao, Discrete Mathematical Structures, New Age International, 2007; ISBN, 8122426697, 9788122426694
2	N Ch S N Iyengar, V M Chandrasekaran, Discrete Mathematics, Vikas Publishers, First Edition, 2003, ISBN :978-8125913627

Assessment and Evaluation

Continuous Internal Evaluation (CIE)	
Components of CIE	Weightage of each component (Marks)
- CIE 1-A (Pen and Paper)	20 out of 70
- CIE 2	25 out of 70
- CIE 3-A (Tutorial)	5 out of 70
- CIE 3-B (Coding Activity)	20 out of 70

Semester End Exams (SEE)	
Mode of Exam	Pen and Paper
Weightage of exam	30

Course Outcomes- Programme Outcomes Matrix

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	3	2	1	1	2	-	-	-	-	-	-	-
CO2	1	1	1	2	3	-	-	-	-	-	-	-
CO3	2	1	2	1	1	-	-	-	-	-	-	-
CO4	3	2	2	3	2	-	-	-	-	-	-	-

Rubrics for evaluation of CIE

Total Marks		20
CIE-1 Components		Rubrics for Assessment
CIE-1	Pen and Paper Exam	The test will be conducted for 20 marks and will be evaluated according to the Scheme of Evaluation prepared by the course faculty.

Total Marks		25
CIE-2 Components		Rubrics for Assessment
CIE-2	Theory	The test will be conducted for 25 marks and will be evaluated according to the Scheme of Evaluation prepared by the course faculty.

Total Marks		25
CIE-3 Components		Rubrics for Assessment
CIE-3A		Assessed based on preparation and participation (2), correctness of problem solving (2), and clarity of explanation (1).
CIE-3B	Coding Activity	Written Python Code (10 marks): Correct logic, proper syntax, readability, and brief comments. Running Code with Different Data (10 marks): Program runs without errors and produces correct output for multiple inputs.

5 marks(CIE-3A)

Criteria	Marks	Description
Preparation & Participation	2	Regular attendance, active involvement in tutorial sessions, preparedness with assigned problems.
Problem Solving Accuracy	2	Correct method and logical steps while solving tutorial problems.
Explanation & Clarity	1	Clear explanation of solution approach and reasoning.
Total	5	

10 marks(CIE-3B)

Criteria	Excellent (9–10)	Good (7–8)	Satisfactory (5–6)	Needs Improvement (0–4)
Logic & Algorithm	Correct and efficient logic; solves the	Correct logic with minor inefficiencies	Basic logic; partially correct	Incorrect or incomplete logic

	problem completely			
Syntax & Structure	Error-free syntax; well-structured code	Minor syntax issues, no major errors	Multiple syntax issues but executable	Code does not execute
Readability & Formatting	Proper indentation, meaningful variable names	Mostly readable, minor formatting issues	Poor naming or inconsistent formatting	Very poor readability
Comments & Explanation	Clear, concise comments explaining logic	Few comments, partially explanatory	Minimal or unclear comments	No comments

10 marks(CIE-3B)

Criteria	Excellent (9–10)	Good (7–8)	Satisfactory (5–6)	Needs Improvement (0–4)
Execution	Runs smoothly without errors	Runs with very minor issues	Runs with some errors	Fails to run
Input Handling	Handles multiple and varied inputs correctly	Handles most inputs correctly	Limited input handling	Incorrect input handling
Output Correctness	Accurate output for all test cases	Mostly correct outputs	Partially correct outputs	Incorrect outputs

Robustness	Handles edge cases effectively	Handles some edge cases	Rarely handles edge cases	No edge case handling
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Signature of the Course Faculty